

Q 1. What do you mean by image processing. Explain the various steps of image processing with the help of a block diagram.

Ans: Image processing deals with developing a system that performs operations on a digital image. The objective of image processing is to transform a given image into a new image, so that it can be suitable for a particular application.

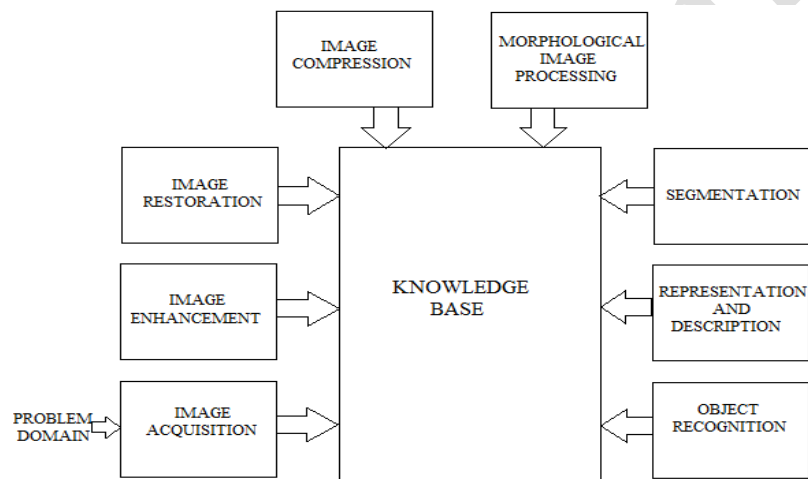
The different steps of image processing are given below.

Image Acquisition: This step aims to obtain the digital image of an object.

Image Enhancement: This step is used to improve the quality of the image, so that the analysis of the image is reliable.

Image Restoration: It deals with improving the appearance of an image. Restoration techniques are based on mathematical or probabilistic models of image degradation.

Image Compression: It is used to reduce the storage requirements to save an image or to reduce bandwidth required to transmit an image.



Morphological Image Processing: It deals with tools for extracting image components that are useful in the representation and description of image.

Segmentation: This step divides the image into its constituent parts or objects or sub regions and extracts the regions that are necessary for further analysis.

Representation & Description: It is the process where the features of the object are extracted (description) and represented using various schemes.

Recognition : It is the process that assigns a label to an object based on its descriptors.

Q 2. Explain various gray level slicing techniques. What is contrast stretching.

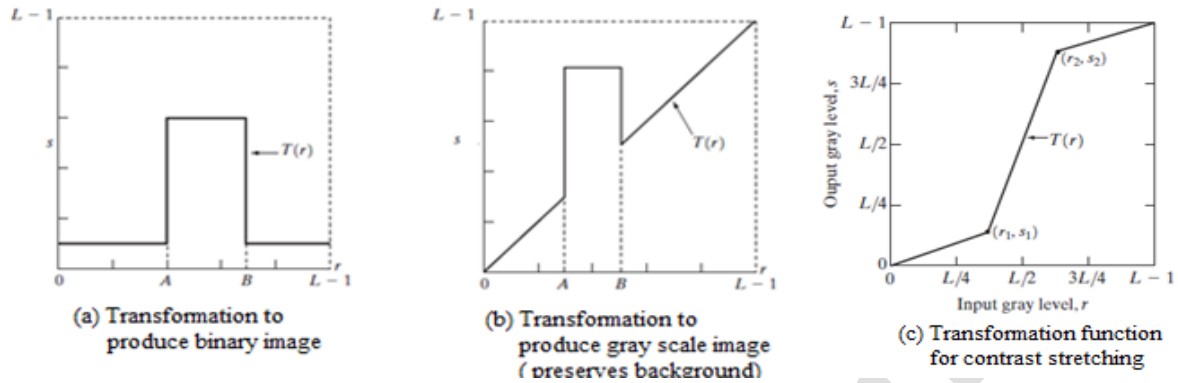
Ans: Gray level slicing is required to highlight a specific range of gray levels in an image. Two techniques of gray level slicing are –

- To display a high value for all gray levels in the specific range of interest and a low value for all other gray levels. This transformation produces a binary image.
- Brightens the desired range of gray levels but preserves the background and gray level tonalities in the image. This transformation produces a gray scale image.

Contrast Stretching: It is a simple piecewise linear transformation function which is used to increase the dynamic range of the gray levels in the image being processed. The location of points (r_1, s_1) and (r_2, s_2) in figure (c) controls the shape of the transformation

function. If $r_1 = s_1$ and $r_2 = s_2$, it is a linear function that produces no change in the gray level. If $r_1 = r_2$, $s_1 = 0$ and $s_2 = L - 1$, the transformation becomes a thresholding function that creates a binary image.

In general, $r_1 \leq r_2$ and $s_1 \leq s_2$ is assumed so that the function is single valued and monotonically increasing.

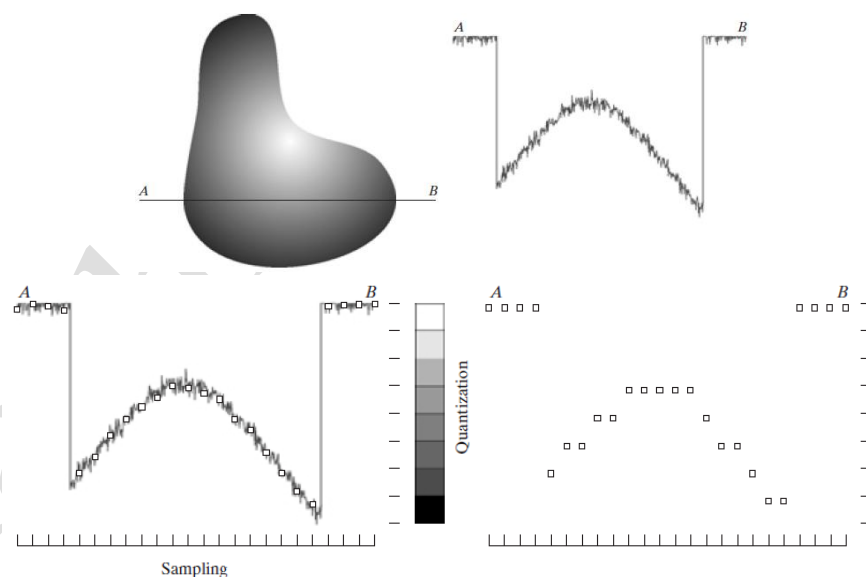


Q 3. Explain sampling and quantization. Explain the effects of reducing sampling and quantization.

Ans: To create a digital image, continuous sensed data needs to be converted into digital form, which involves two processes: Sampling and Quantization.

Sampling is the process in which the coordinate values of an image are digitized.

Quantization is the process in which the amplitude values of an image are digitized.



Q 4. Explain homomorphic filtering.

Ans: Appearance of an image can be improved by using the illumination-reflectance model by simultaneous gray level range compression and contrast enhancement.

Illumination component of an image is generally characterized by slow spatial variation while the reflectance component of an image tends to vary abruptly. These characteristics lead to associating the low frequencies of the Fourier transform of the natural log of an image with illumination and high frequencies with reflectance. Even though these assumptions are

approximation at best, a good deal of control can be gained over the illumination and reflectance components with a homomorphic filter.

Homomorphic filtering is a method in which the illumination and reflectance components can be filtered individually. The basic model is as shown below.

An image $f(x, y)$ can be expressed as the product of illumination and reflectance components.

$$f(x, y) = i(x, y) \cdot r(x, y) \quad (1)$$

As, the fourier transform of the product of two functions is not separable, i.e.

$$F\{f(x, y)\} \neq F\{i(x, y)\} \cdot F\{r(x, y)\}$$

Therefore, equation (1) can't be used directly to operate separately on the frequency components of illumination and reflection.

$$\text{Let, } z(x, y) = \ln [f(x, y)] = \ln [i(x, y)] + \ln [r(x, y)]$$

$$\text{So, } F\{z(x, y)\} = F\{\ln [i(x, y)]\} + F\{\ln [r(x, y)]\}$$

$$\text{Or, } Z(u, v) = F_i(u, v) + F_r(u, v)$$

If, we process $Z(u, v)$ by means of a filter function $H(u, v)$, then,

$$S(u, v) = H(u, v) \cdot Z(u, v)$$

$$S(u, v) = H(u, v) F_i(u, v) + H(u, v) F_r(u, v)$$

where, $S(u, v)$ is the fourier transform of the result. In the spatial domain,

$$s(x, y) = F^{-1}\{S(u, v)\} = F^{-1}\{H(u, v) F_i(u, v)\} + F^{-1}\{H(u, v) F_r(u, v)\}$$

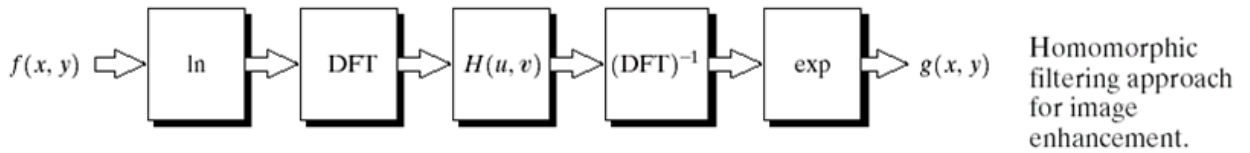
$$\text{Let, } i'(x, y) = F^{-1}\{H(u, v) F_i(u, v)\} \quad \text{and} \quad r'(x, y) = F^{-1}\{H(u, v) F_r(u, v)\}$$

$$s(x, y) = i'(x, y) + r'(x, y)$$

Therefore, enhanced image $g(x, y)$ is given by – $g(x, y) = e^{s(x, y)} = e^{i'(x, y)} e^{r'(x, y)}$

$$g(x, y) = i_0(x, y) \cdot r_0(x, y)$$

where, $i_0(x, y) = e^{i'(x, y)}$ and $r_0(x, y) = e^{r'(x, y)}$ are the illumination and reflectance components of the output image.



In homomorphic filtering, the homomorphic filter function $H(u, v)$ is operated after the separation of the illumination and reflectance components as described above.

Q 5. Explain histogram equalization. Derive the formula to perform histogram equalization.

Ans: **Histogram equalization** is a common technique for enhancing the appearance of images. Suppose we have an image which is predominantly dark. Then its histogram would be skewed towards the lower end of the grey scale and all the image detail is compressed into the dark end of the histogram. If we could 'stretch out' the grey levels at the dark end to produce a more uniformly distributed histogram then the image would become much cleaner.

Histogram equalization involves finding a grey scale transformation function that creates an output image with a uniform histogram.

Assume our grey levels are continuous and have been normalized to lie between 0 and 1. We must find a transformation T that maps grey values r in the input image F to gray values $s = T(r)$ in the transformed image F .

It is assumed that T is single valued & monotonically increasing and $0 \leq T(r) \leq 1$ for $0 \leq r \leq 1$. The inverse transformation from s to r is given by, $r = T^{-1}(s)$. If one takes the histogram for

the input image and normalizes it so that the area under the histogram is 1, we have a probability distribution for grey levels in the input image $P_r(r)$.

From probability theory, $P_s(S) = P_r(r)[dr/ds]$ where, $r = T^{-1}(s)$.

Consider the transformation, $s = T(r) = \int P_r(\omega)d\omega$

This is the cumulative distribution function of r . Using this definition of T we see that the derivative of s with respect to r is $ds/dr = P_r(r)$

Substituting this back into the expression for P_s , we get $P_s(S) = P_r(r)[1/P_r(r)] = 1$ for all s , where $0 \leq s \leq 1$. Thus, $P_s(s)$ is now a uniform distribution function.

Q 6. Explain the 4, 8 and m connectivity of pixels with the help of an example.

Ans: Two pixels 'p' and 'q' are said to be connected if they are neighbors and their gray levels satisfy a specified condition/criterion of similarity (or equal).

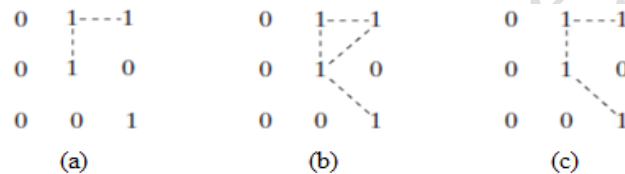
Let, 'V' be the set of gray levels used to define connectivity.

4-Connectivity: Two pixels 'p' and 'q' with values from 'V' are 4-connected if $q \in N_4(p)$.

8-Connectivity: Two pixels 'p' and 'q' with values from 'V' are 8-connected if $q \in N_8(p)$.

m-Connectivity: Two pixels 'p' and 'q' with values from 'V' are m-connected if

(i) $q \in N_4(p)$. (ii) $q \in N_D(p)$ and set $N_4(p) \cap N_4(q) = \emptyset$.



Q 7. What are the components of an image processing system.

Ans: A general image processing system is composed of the following components:

Image Sensors: These are used for image acquisition. To acquire a digital image, sensors that are sensitive to the light energy radiated by the object, converts light energy into voltage proportional to the intensity of the light. Then, digitizer is used to convert output of sensors into digital form.

Specialized Image Processing Hardware: It consists of digitizer and hardware that is capable of performing arithmetic and logical operations in parallel on entire image. It reduces the processing time and thus, speed of the operation is increased.

Image Processing Software: Software for image processing consists of specialized modules that performs specific tasks. A well designed package also includes the capability for the user to write codes and to integrate modules and general purpose software commands from atleast one computer language. Example – MATLAB.

Computer: The computer in an image processing system is a general-purpose computer and can range from a PC to a supercomputer. In dedicated applications, sometimes specially designed computers are used to achieve a required level of performance. In these systems, almost any well-equipped PC-type machine is suitable for offline image processing tasks.

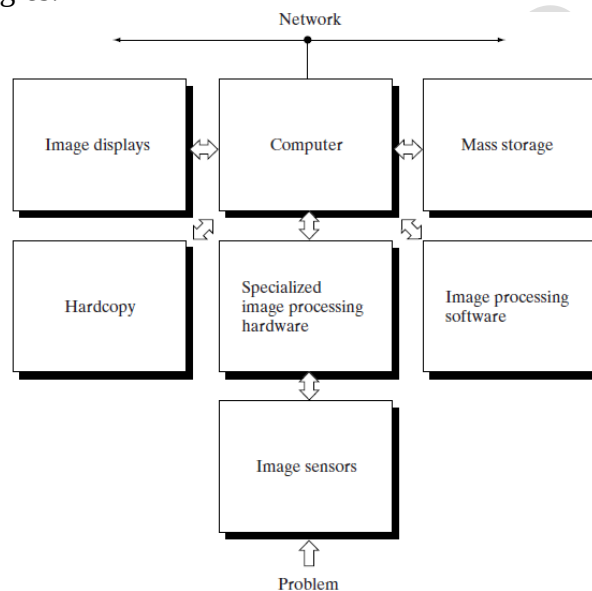
Mass Storage: Mass storage capability is a must in image processing applications. An image of size 1024*1024 pixels, in which the intensity of each pixel is an 8-bit quantity, requires one megabyte of storage space if the image is not compressed. Digital storage for image processing applications falls into three principal categories: (1) short-term storage for use during processing, (2) on-line storage for relatively fast recall, and (3) archival storage,

characterized by infrequent access. Storage is measured in bytes, kilobytes, mega-bytes, giga-bytes and tera-bytes.

Image Displays: Image displays in use today are mainly color (preferably flat screen) TV monitors. Monitors are driven by the outputs of image and graphics display cards that are an integral part of the computer system.

Hardcopy: Hardcopy devices for recording images include laser printers, film cameras, heat-sensitive devices, inkjet units, and digital units, such as optical and CD-ROM disks. Film provides the highest possible resolution, but paper is the obvious medium of choice for written material. For presentations, images are displayed on film transparencies or in a digital medium if image projection equipment is used.

Network: Networking is almost a default function in any computer system in use today. Because of the large amount of data inherent in image processing applications, the key consideration in image transmission is bandwidth. In dedicated networks, this typically is not a problem, but communications with remote sites via the Internet are not always as efficient. Fortunately, this situation is improving quickly as a result of optical fiber and other broadband technologies.



Q 8. Obtain the digital negative of the following 8 - bit sub image.

139	205	105
141	252	99
201	15	76

Ans: As, the given sub image is of 8-bit, therefore, $L = 256$.

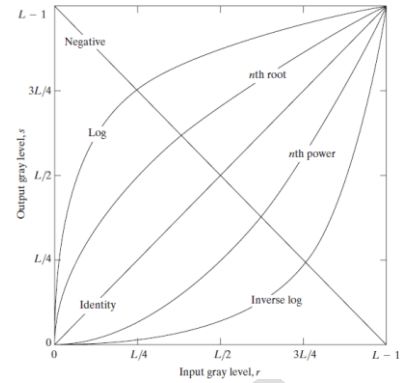
Using the image negative transformation given by $s = L - 1 - r$,

Image Negative of the given sub - image is -

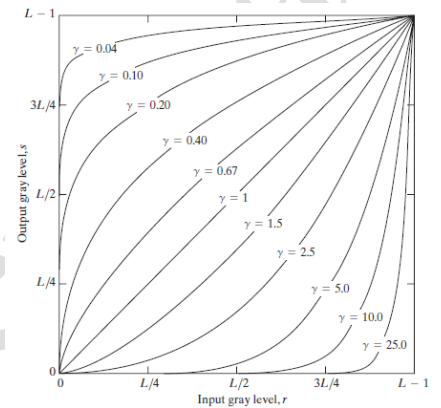
116	50	150
114	3	156
54	240	179

Q 9. Explain the power law transformation and log transformation. What are their applications. Also, explain gamma correction.

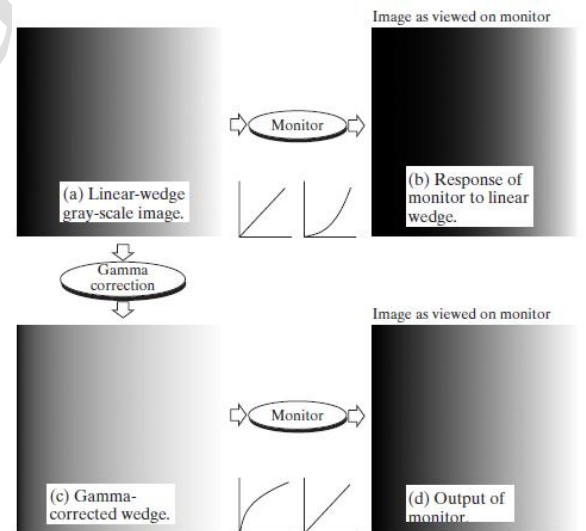
Ans: Log Transformation: The general form of log transformation is: $s = c \log(1 + r)$, where, c is a constant and $r \geq 0$. Log transformation maps a narrow range of low gray – level values in the input image into a wider range of output levels. Also, this transformation expands the values of dark pixels in an image while compresses the higher level values. The log function has the important characteristic that it compresses the dynamic range of the images with large variations in pixel values.



Power Law Transformation: The basic form of power law transformation is: $s = cr^\gamma$, where, ' c ' and ' γ ' are positive constants. Power – law transformation with fractional values of γ map a narrow range of dark input values into a wider range of output values and vice – versa for higher values of input levels. In power – law transformation, a family of transformation curves can be obtained simply by varying γ . Curves obtained by selecting $\gamma > 1$ expands the upper range of gray levels while lower and mid – range of gray – levels are suppressed. Curves obtained by selecting $\gamma < 1$ expands the lower range of gray levels while mid and upper – range of gray – levels are suppressed.



Gamma Correction: A variety of devices used for image capture, printing, and display respond according to a power law. By convention, the exponent in the power-law equation is referred to as *gamma*. The process used to correct this power-law response phenomena is called *gamma correction*. For example, cathode ray tube (CRT) devices have an intensity-to-voltage response that is a power function, with exponents varying from approximately 1.8 to 2.5. Such display systems would tend to produce images that are darker than intended and the output of the monitor appears darker than the input. In this case, for gamma correction, the input image is preprocessed before applying it as input for monitor by performing the transformation $s = r^{\frac{1}{2.5}} = r^{0.4}$. When input into the same monitor, this gamma corrected input produces an output image that is close in appearance to the original image. If the requirement is to display the image accurately on a computer screen, then gamma correction is important. Images without gamma correction may look either bleached out or too dark.



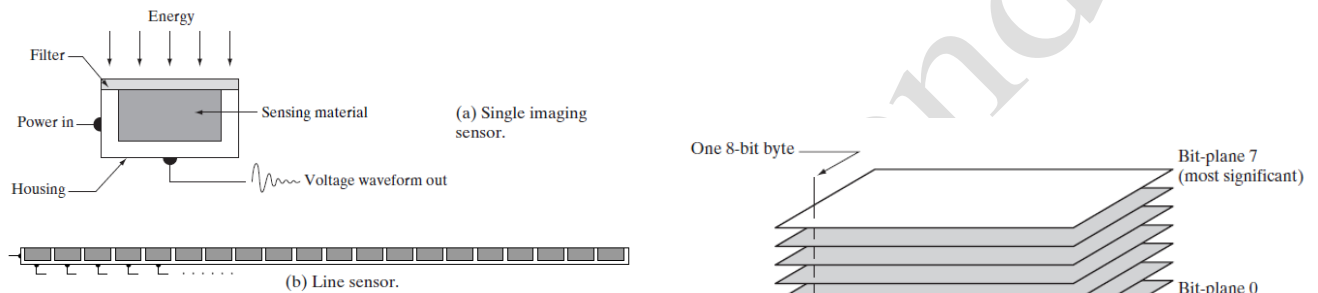
Q 10. Write short note on the following:

Ans: Image acquisition using sensor strips:

Image acquisition using sensor strips is used more frequently than single sensors. It consists of an in-line arrangement of sensors in the form of a sensor strip as shown in the figure. The strip provides imaging elements in one direction. Motion perpendicular to the strip provides

imaging in the other direction. This is the type of arrangement used in most flat bed scanners. Sensing devices with 4000 or more in-line sensors are possible. In-line sensors are used routinely in airborne imaging applications, in which the imaging system is mounted on an aircraft that flies at a constant altitude and speed over the geographical area to be imaged. One-dimensional imaging sensor strips that respond to various bands of the electromagnetic spectrum are mounted perpendicular to the direction of flight. The imaging strip gives one line of an image at a time, and the motion of the strip completes the other dimension of a two-dimensional image. Lenses or other focusing schemes are used to project the area to be scanned onto the sensors.

Sensor strips mounted in a ring configuration are used in medical and industrial imaging to obtain cross-sectional (“slice”) images of 3-D objects. The output of the sensors must be processed by reconstruction algorithms whose objective is to transform the sensed data into meaningful cross-sectional images.



Bit Plane Slicing:

Sometimes in image processing, it is desired to highlight the contribution made to total image appearance by specific bits. Suppose that each pixel in an image is represented by 8 bits. Imagine that the image is composed of eight 1-bit planes, ranging from bit-plane 0 for the least significant bit to bit plane 7 for the most significant bit. In terms of 8-bit bytes, plane 0 contains all the lowest order bits in the bytes comprising the pixels in the image and plane 7 contains all the high-order bits. Figure shows the various bit planes for the image. The higher-order bits (especially the top four) contain the majority of the visually significant data. The other bit planes contribute to more subtle details in the image. Separating a digital image into its bit planes is useful for analyzing the relative importance played by each bit of the image, a process that aids in determining the adequacy of the number of bits used to quantize each pixel. Also, this type of decomposition is useful for image compression. In terms of bit-plane extraction for an 8-bit image, the binary image for bit-plane 7 can be obtained by processing the input image with a thresholding gray-level transformation function that (1) maps all levels in the image between 0 and 127 to one level (for example, 0); and (2) maps all levels between 129 and 255 to another (for example, 255). The bit plane slicing for a 3 X 3 subimage is shown below.

Original Image			Binary Representation of Image		
0	3	5	000	011	101
1	7	4	001	111	100
2	6	5	010	110	101

Bit Plane 2			Bit Plane 1			Bit Plane 0		
0	0	1	0	1	0	0	1	1
0	1	1	0	1	0	1	1	0
0	1	1	1	1	0	0	0	1

Q 11. What do you understand by histogram and normalized histogram of an image. What information about the image can be obtained by analyzing the histogram. An image segment is shown below. Let, V be the set of gray level values used to define connectivity in the image. Compute D_4 , D_8 and D_m distances between pixels 'p' and 'q' for :

(i) $V = \{2,3\}$

(ii) $V = \{2,6\}$

2 (p)	3	2	6	1
6	2	3	6	2
5	3	2	3	5
2	4	3	5	2
4	5	2	3	3 (q)

Ans: Histogram: The histogram of a digital image with gray levels in the range $[0, L-1]$ is a discrete function $h(r_k)=n_k$, where r_k is the k th gray level and n_k is the number of pixels in the image having gray level r_k . Normalized histogram can be obtained by dividing each of its values by the total number of pixels in the image, denoted by n . Thus, a normalized histogram is given by $p(r_k)=n_k/n$, for $k=0, 1,...,L-1$. $p(r_k)$ gives an estimate of the probability of occurrence of gray level r_k . The sum of all components of a normalized histogram is equal to 1. In the dark image, the components of the histogram are concentrated on the low (dark) side of the gray scale. Similarly, the components of the histogram of the bright image are biased toward the high side of the gray scale. An image with low contrast has a narrow histogram and will be centered toward the middle of the gray scale. For a monochrome image this results in dull, washed-out gray look. The components of the histogram in the high-contrast image cover a broad range of the gray scale and the distribution of pixels is approximately uniform. An image whose pixels tend to occupy the entire range of possible gray levels and tend to be distributed uniformly, will have an appearance of high contrast and will exhibit a large variety of gray tones. The net effect will be an image that shows a great deal of gray-level detail and has high dynamic range.

As, D_4 and D_8 distance depends only on the coordinates of pixels and not on the gray levels of pixels. Therefore,

Coordinates of p $(x, y) = (0, 0)$ and Coordinates of q $(s, t) = (4, 4)$.

$D_4(p, q) = |x - s| + |y - t| = |0 - 4| + |0 - 4| = 4 + 4 = 8$ units.

$D_8(p, q) = \max(|x - s|, |y - t|) = \max(|0 - 4|, |0 - 4|) = 4$ units.

D_m distance between two pixels depends on the values of the pixels along with the path and also on the values of their neighbours.

(i) If $V = \{2, 3\}$.

2 (p)	3	2	6	1
6	2	3	6	2
5	3	2	3	5
2	4	3	5	2
4	5	2	3	3 (q)

D_m distance = 7 pixels.

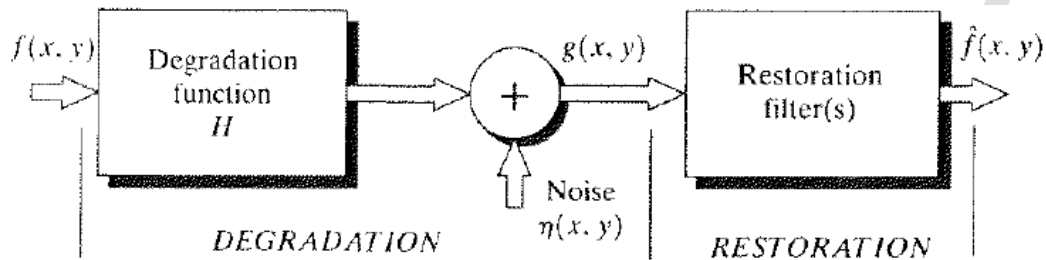
(ii) If $V = \{2, 6\}$.

There is no path through which 'p' and 'q' are connected. Therefore, D_m distance for this case can't be computed.

2 (p)	3	2	6	1
6	2	3	6	2
5	3	2	3	5
2	4	3	5	2
4	5	2	3	3 (q)

Q 12. Explain image degradation and restoration model along with mathematical equations in spatial and frequency domain.

Ans: Model of Image Degradation / Restoration Process



As shown in the figure, the degradation process is modeled as a degradation function that operates on an input image $f(x, y)$, with an additive noise term $\eta(x, y)$ to produce a degraded image $g(x, y)$.

If $g(x, y)$ is given, the objective of restoration is to obtain an estimate $\hat{f}(x, y)$ of the original image, with some knowledge about the degradation function H and additive noise term $\eta(x, y)$.

If H is a linear, position – invariant process, then the degraded image in spatial domain is given by -

$$g(x, y) = h(x, y) * f(x, y) + \eta(x, y)$$

where, $h(x, y)$ is the spatial representation of the degradation function and $*$ denotes the convolution operation.

By using the convolution theorem, which states that convolution in the spatial domain is equal to the multiplication in the frequency domain, an equivalent frequency domain representation of the above equation is given as -

$$G(u, v) = H(u, v) F(u, v) + N(u, v)$$

where, $G(u, v)$, $H(u, v)$, $F(u, v)$ and $N(u, v)$ are the fourier transforms of $g(x, y)$, $h(x, y)$, $f(x, y)$ and $\eta(x, y)$ respectively.

For noise analysis, we are assuming that the degradation function is an identity operator (equals to 1), i.e. it doesn't have any effect on the input image.

Q 13. Write a short note on image averaging.

Ans: **Image Averaging:** It is used for noise reduction. It is achieved by adding the corresponding pixel image of all the noisy images and then dividing them with the number of noisy images to get the averaged intensity. If the noise at each pixel is uncorrelated, then, it gets cancelled out during averaging. Better results can be obtained as the numbers of noisy images are increased.

$$g(x, y) = f(x, y) + n(x, y) ;$$

where, $g(x, y)$, $f(x, y)$ and $n(x, y)$ are the noisy image, original image and noise respectively.

$$\bar{g}(x, y) = \frac{1}{K} \sum_{i=1}^K g_i(x, y)$$

$$E \{ \bar{g}(x, y) \} = f(x, y)$$

$$\sigma_{\bar{g}(x, y)}^2 = \frac{1}{K} \sigma_{\eta(x, y)}^2$$

$$\sigma_{\bar{g}(x, y)} = \frac{1}{\sqrt{K}} \sigma_{\eta(x, y)}$$

As K increases, the variability (noise) of pixel values at each location (x, y) decreases. Also, $E \{ \bar{g}(x, y) \} = f(x, y)$, therefore, $\bar{g}(x, y)$ approaches $f(x, y)$ as the number of noisy images used in the averaging process increases.

Q 14. Write a note on noise models in image restoration. Describe Wiener Filter and Inverse Filtering.

Ans: The principal source of noise in digital images arises during image acquisition and transmission. The performance of imaging sensors is affected by a variety of factors, such as environmental conditions during image acquisition and by the quality of the sensing elements themselves. Images are corrupted during transmission principally due to interference in the channels used for transmission. Since main sources of noise presented in digital images are resulted from atmospheric disturbance and image sensor circuitry, following assumptions can be made:

- The noise model is spatial invariant, i.e., independent of spatial location.
- The noise model is uncorrelated with the object function.
-

Some important Noise Probability Density Functions (PDF)/Noise Models

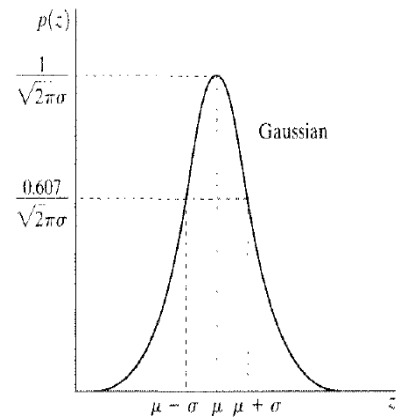
Gaussian/ Normal Noise Model:

The PDF of a Gaussian random variable, z is given by –

$$p(z) = \frac{1}{\sqrt{2\pi}\sigma} e^{-(z-\mu)^2/2\sigma^2}$$

where, ‘ z ’ represents gray level, ‘ μ ’ is the mean of average values of ‘ z ’ and ‘ σ ’ is its standard deviation. Also, the square of standard deviation σ^2 is known as variance of ‘ z ’.

For Gaussian PDF, approximately 70% of the values lies in the range $[(\mu - \sigma), (\mu + \sigma)]$ and about 95% will be in the range $[(\mu - 2\sigma), (\mu + 2\sigma)]$.

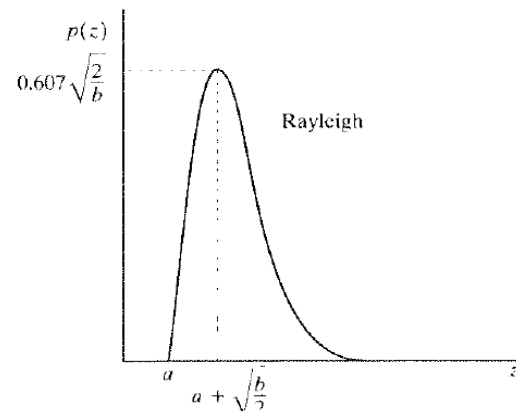


Rayleigh Noise:

The PDF of Rayleigh noise is given by –

$$p(z) = \begin{cases} \frac{2}{b}(z - a)e^{-\frac{(z-a)^2}{b}} & \text{for } z \geq a \\ 0 & \text{for } z < a \end{cases}$$

The mean and variance are given by –



$$\mu = a + \sqrt{\pi b/4}$$

$$\sigma^2 = \frac{b(4 - \pi)}{4}$$

Erlang (Gamma) Noise:

The PDF of Erlang Noise is given by –

$$p(z) = \begin{cases} \frac{a^b z^{b-1}}{(b-1)!} e^{-az} & \text{for } z \geq 0 \\ 0 & \text{for } z < 0 \end{cases}$$

where, $a > 0$ and b is a positive integer.

The mean and variance are given by –

$$\mu = b/a \quad \text{and} \quad \sigma^2 = b/a^2$$

Exponential Noise:

Exponential PDF is a special case of Erlang PDF with $b = 1$.

The PDF of Exponential Noise is given by –

$$p(z) = \begin{cases} ae^{-az} & \text{for } z \geq 0 \\ 0 & \text{for } z < 0 \end{cases}$$

where, $a > 0$.

The mean and variance are given by –

$$\mu = 1/a \quad \text{and} \quad \sigma^2 = 1/a^2$$

Uniform Noise:

The PDF of uniform noise is given by –

$$p(z) = \begin{cases} \frac{1}{b-a} & \text{if } a \leq z \leq b \\ 0 & \text{otherwise} \end{cases}$$

The mean and variance are given by –

$$\mu = (a + b)/2 \quad \text{and} \quad \sigma^2 = (b - a)^2/12$$

Impulse (Salt and Pepper) Noise:

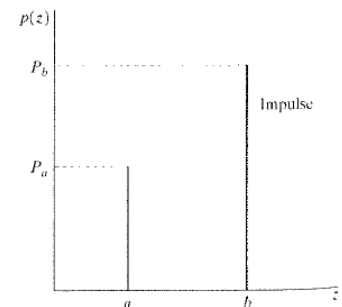
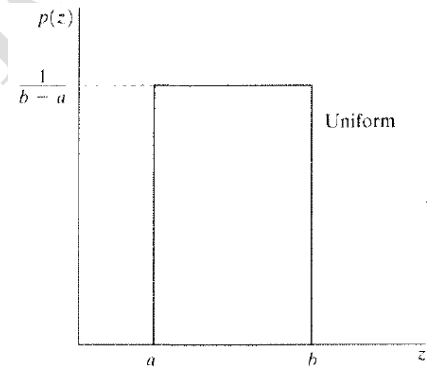
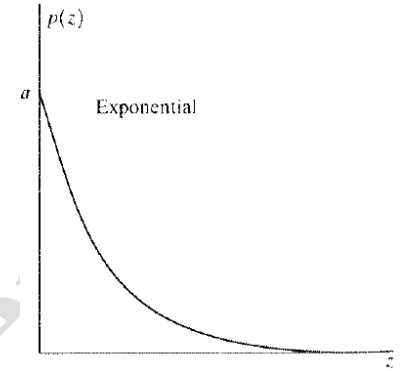
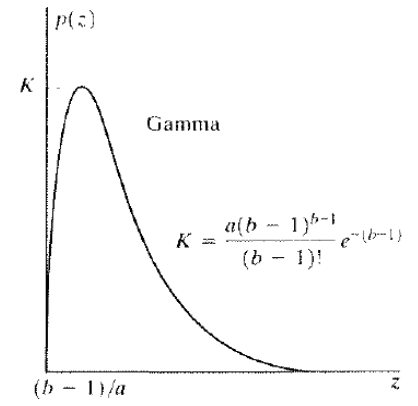
The PDF of bipolar impulse noise is given by –

$$p(z) = \begin{cases} P_a & \text{for } z = a \\ P_b & \text{for } z = b \\ 0 & \text{otherwise} \end{cases}$$

If $b > a$, gray-level ‘b’ will appear as a light dot in the image and level ‘a’ will appear like a dark dot. If either P_a or P_b is zero, the impulse noise is called Unipolar.

If neither probability is zero, and especially if they are approximately equal, impulse noise values will resemble salt and pepper granules randomly distributed over the image. For, this reason, Bipolar Impulse Noise is also known as Salt and Pepper Noise. It is also known as Shot and Spike Noise.

Noise impulses can be positive or negative. Usually, image corruption is large compared with the strength of the image signal, impulse noise generally is digitized as extreme values in the image (pure black or white). Thus, **a** and **b** are saturated values or they are equal to the maximum and minimum allowed values in the digitized image. So, negative



impulses appear as **black (pepper)** points and positive impulses appear as **white (salt)** points/noise in an image.

Applications:

Gaussian noise arises in an image due to factors such as electronic circuit noise and sensor noise due to poor illumination and or high temperature.

Rayleigh density is helpful in characterizing noise phenomenon in Range Imaging.

Exponential and **Gamma** densities find application in Laser Imaging.

Impulse noise is found in situations where quick transients, such as faulty switching takes place during imaging.

Uniform density is the least descriptive of practical situations. It is quite useful as basis for numerous random number generators which are useful in simulations.

Q 15. Derive the mask to implement Laplacian in spatial domain.

Ans: From the definition of digital second order derivative,

$$\frac{\partial^2 f}{\partial x^2} = f(x+1, y) + f(x-1, y) - 2f(x, y)$$

$$\frac{\partial^2 f}{\partial y^2} = f(x, y+1) + f(x, y-1) - 2f(x, y)$$

Laplacian of a two dimensional function $f(x, y)$ is given by –

$$\nabla^2 f = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2}$$

$$\nabla^2 f = [f(x+1, y) + f(x-1, y) - 2f(x, y)] + [f(x, y+1) + f(x, y-1) - 2f(x, y)]$$

$$\nabla^2 f = f(x+1, y) + f(x-1, y) + f(x, y+1) + f(x, y-1) - 4f(x, y)$$

Therefore, mask of Laplacian is given by –

0	1	0
1	-4	1
0	1	0

Q 16. Explain high boost filtering and unsharp masking.

Ans: **Unsharp Masking:** It is used for image sharpening in publishing industry. It consists of subtracting a blurred version of an image from the image itself.

$$f_s(x, y) = f(x, y) - \bar{f}(x, y)$$

where, $f(x, y)$ is the input image, $f_s(x, y)$ is the sharpened image obtained by unsharp masking and $\bar{f}(x, y)$ is the blurred version of $f(x, y)$.

High Boost Filtering: It is the generalized form of unsharp masking. A high – boost filtered image f_{hb} at any point (x, y) is defined as –

$$f_{hb}(x, y) = Af(x, y) - \bar{f}(x, y), \quad A \geq 1$$

where, A is known as boost coefficient.

$$f_{hb}(x, y) = (A - 1)f(x, y) + f(x, y) - \bar{f}(x, y)$$

$$f_{hb}(x, y) = (A - 1)f(x, y) + f_s(x, y)$$

Therefore,

$$f_{hb} = \begin{cases} Af(x, y) - \nabla^2 f(x, y); & \text{If the centre coefficient of the Laplacian Mask is negative.} \\ Af(x, y) + \nabla^2 f(x, y); & \text{If the centre coefficient of the Laplacian Mask is positive.} \end{cases}$$

Mask of High – Boost Filtering is given by –

0	-1	0
-1	A+4	-1
0	-1	0

Q 17. Find the output of applying ‘AND’ and ‘OR’ operations on the given 3×3 subimages.

3	6	4
1	3	2
5	4	2

Image A

5	1	3
3	5	6
7	0	2

Image B

Ans: Image A =

011	110	100
001	011	010
101	100	010

Image B =

101	001	011
011	101	110
111	000	010

Image A **AND** Image B =

001	000	000
001	001	010
101	000	010

1	0	0
1	1	2
5	0	2

Image A **OR** Image B =

111	111	111
011	111	110
111	100	010

7	7	7
3	7	6
7	4	2

Q 18. The following matrix defines a 5×5 image $f(x, y)$. Suppose smoothing is done to the image using 3×3 neighborhood in the spatial domain. Then, what will be the new value of $f(2, 2)$ using –
 (i) Box Filter. (ii) 100th Percentile Filter. (iii) 50th Percentile Filter.
 (iv) 0th Percentile Filter.

2	3	2	4	5
1	3	5	4	5
2	1	2	7	6
3	6	5	6	4
3	5	6	4	7

Image $f(x, y)$

Ans: Matrix shows the given image and area occupied by 3×3 filter is shown in colored part.

2	3	2	4	5
1	3	5	4	5
2	1	2 f(2,2)	7	6
3	6	5	6	4
3	5	6	4	7

- (i) Response of Box Filter = Mean $\{3,5,4,1,2,7,6,5,6\} = 4.33$
 $f(2,2) = 4$ (After Quantization)
- (ii) Response of 100th Percentile Filter = Max $\{3,5,4,1,2,7,6,5,6\} = 7$
 $f(2,2) = 7$
- (iii) Response of 50th Percentile Filter = Median $\{3,5,4,1,2,7,6,5,6\} = 5$
 $f(2,2) = 5$
- (iv) Response of 0th Percentile Filter = Min $\{3,5,4,1,2,7,6,5,6\} = 1$
 $f(2,2) = 1$

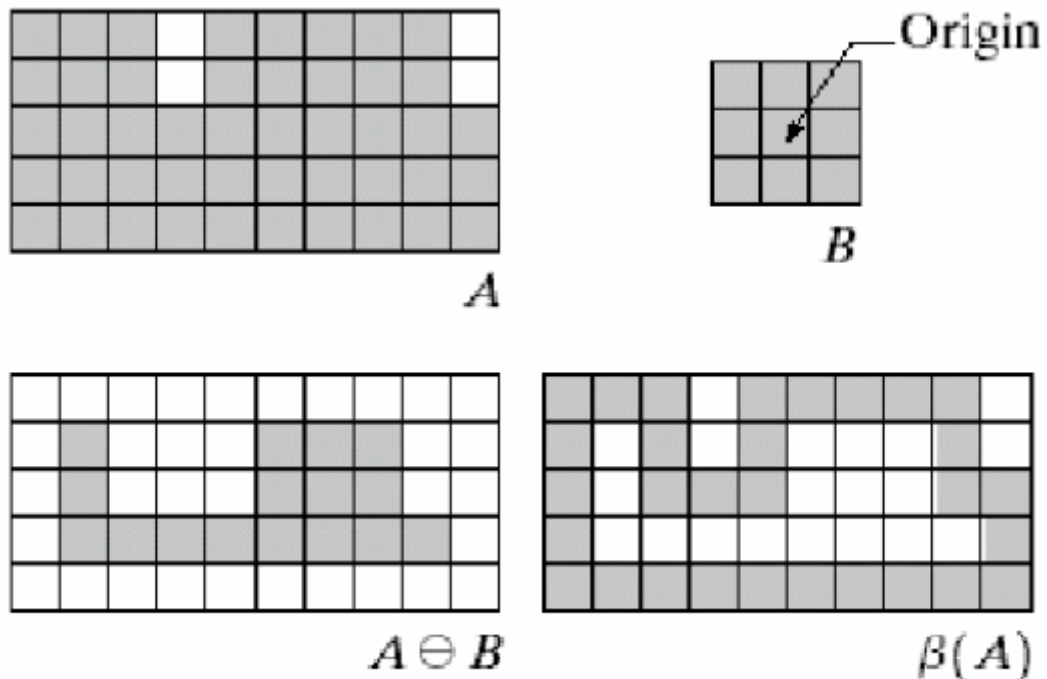
Q 19. What do you mean by morphology. Discuss any one morphological algorithm in detail.

Ans: Morphology is a tool for extracting image components that are useful in the representation and description of region shape, such as boundaries, skeletons and the convex hull.

Boundary Extraction: The boundary of a set A, denoted by $\beta(A)$ can be obtained by first eroding A by B and then performing the set difference between A and its erosion.

$$\beta(A) = A - (A \ominus B)$$

where, B is a suitable structuring element. Figure shows the implementation of boundary extraction algorithm.



Q 20. Explain the following:

Ans: (i) **Convex Hull:** The convex hull H of an arbitrary set S is the smallest convex set containing S . The set difference $H - S$ is called the convex deficiency of S . Morphological algorithm for obtaining convex hull $C(A)$ of a set A is given below.

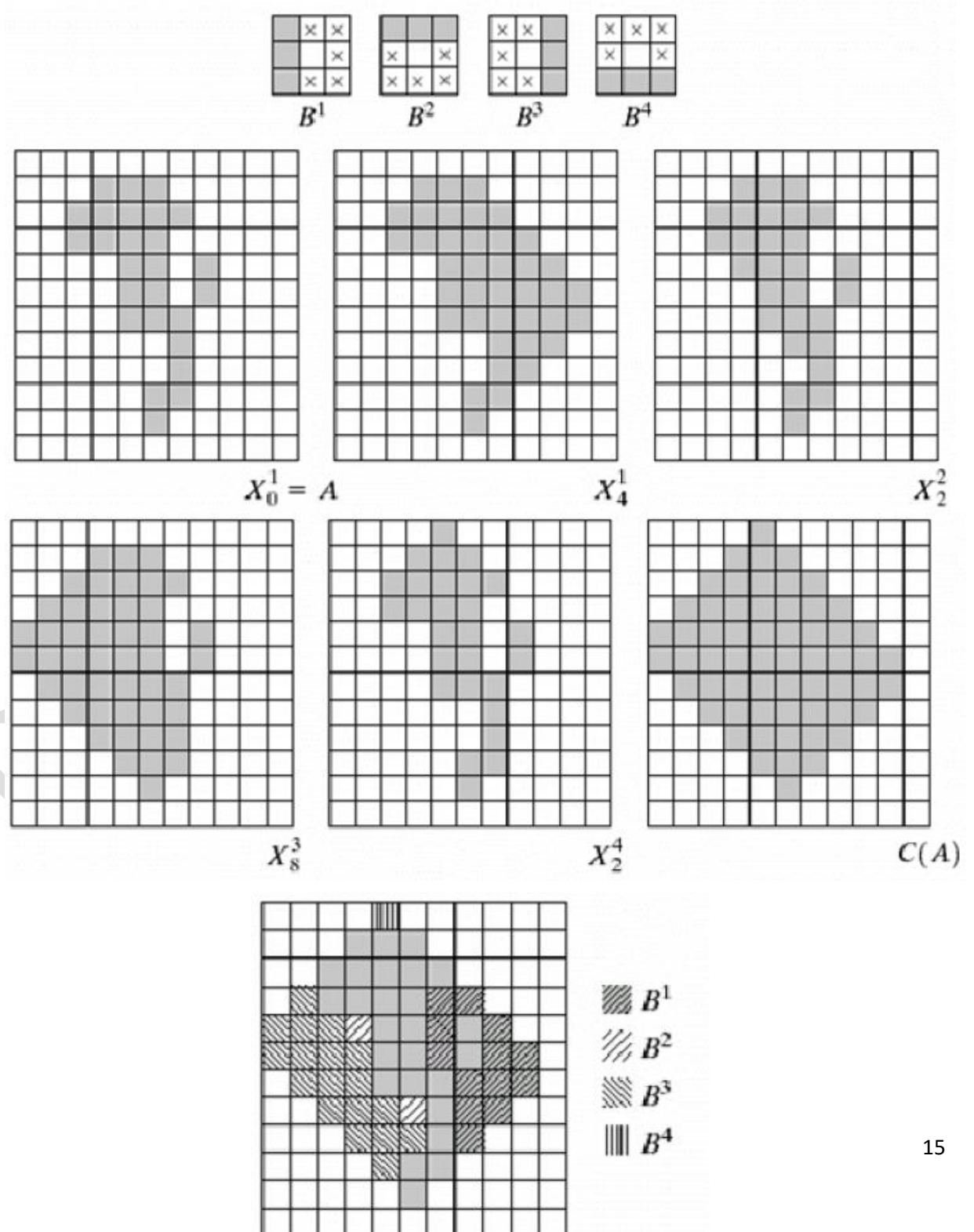
Let, B^1, B^2, B^3 and B^4 represents the four structuring elements as shown in the figure. The procedure consists of implementing the equation

$$X_k^i = (X_{k-1} \circledast B^i) \cup A, \quad i = 1, 2, 3, 4, \quad k = 1, 2, \dots, \quad X_0^i = A$$

Now, let, $D^i = X_{conv}^i$, where “conv” indicates convergence in the sense that $X_k^i = X_{k-1}^i$

Then, the convex hull of A is –

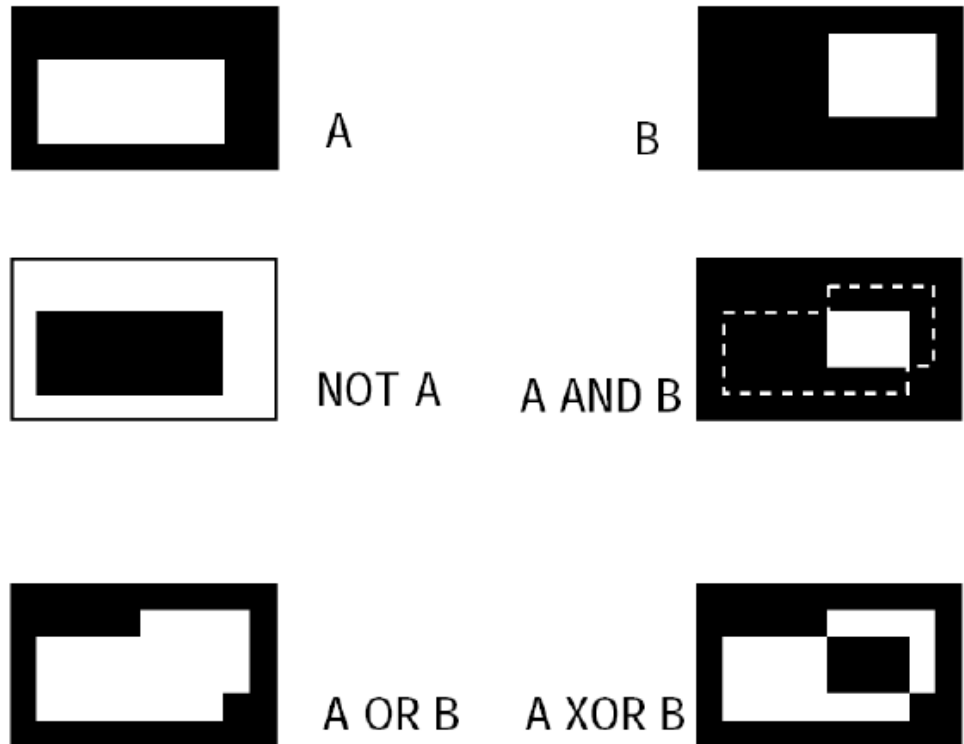
$$C(A) = \bigcup_{i=1}^4 D^i$$



- (ii) **Logic operations involving binary images:** The principle logic operations used in image processing are AND, OR and NOT (Complement). These operations are **functionally complete** because they can be combined to form any other logic operation.

Logic operations are performed on a pixel by pixel basis between corresponding pixels of two or more images (except NOT, which operates on pixels of a single image). The result of applying AND operation will be '1', if the corresponding pixels in the two input images are '1'. The result of applying OR operation will be '1', if the any one of the corresponding pixels in the two input images are '1'. The results of applying these logical operations are shown in the figure below.

Pixel-wise combination of images (AND, OR, NOT, XOR)



Q 21. What do you mean by thinning and thickening of an image. Discuss the method for thinning of an image.

Ans: **Thinning:**

The thinning of a set A by a structuring element B :

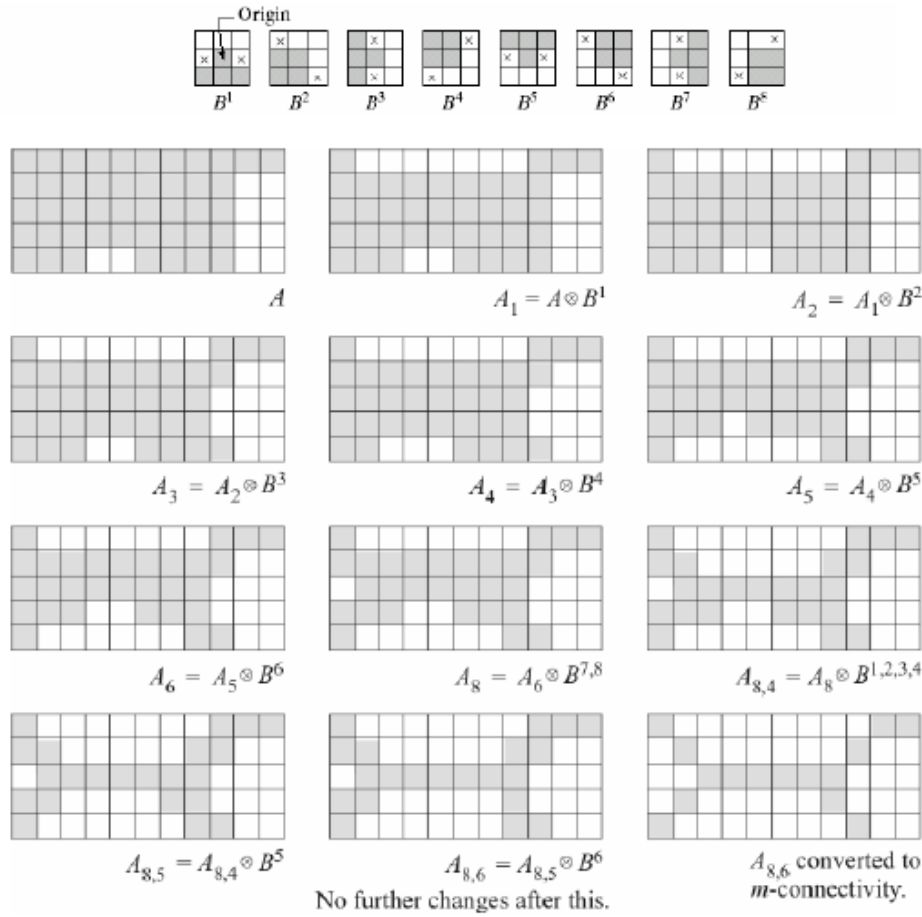
$$A \otimes B = A - (A \circledast B) = A \cap (A \circledast B)^c$$

Symmetric thinning: sequence of structuring elements,

$$\{B\} = \{B^1, B^2, B^3, \dots, B^n\},$$

where B^i is a rotated version of B^{i-1}

$$A \otimes \{B\} = ((\dots((A \otimes B^1) \otimes B^2) \dots) \otimes B^n)$$



Thickening:

Thickening is the morphological dual of thinning and is defined by

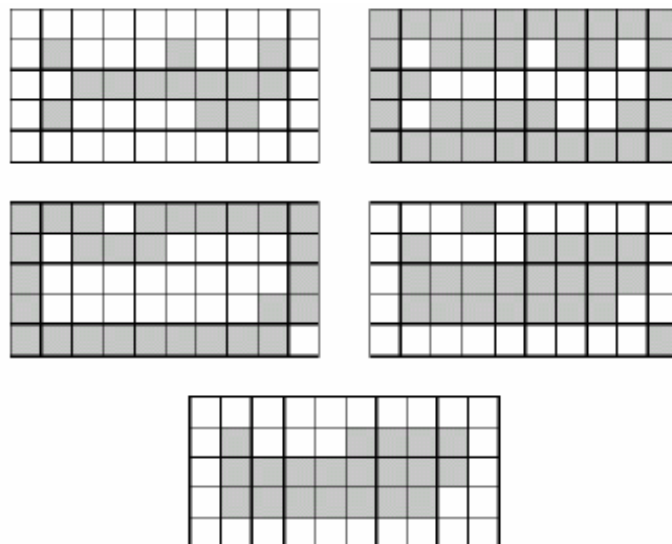
$$A \odot B = A \cup (A \otimes B),$$

where B is a structuring element

Similar to thinning...

$$A \odot \{B\} = ((\dots((A \odot B^1) \odot B^2) \dots) \odot B^n)$$

Structuring elements for thickening are similar to those of figure, but with all 1's and 0's interchanged



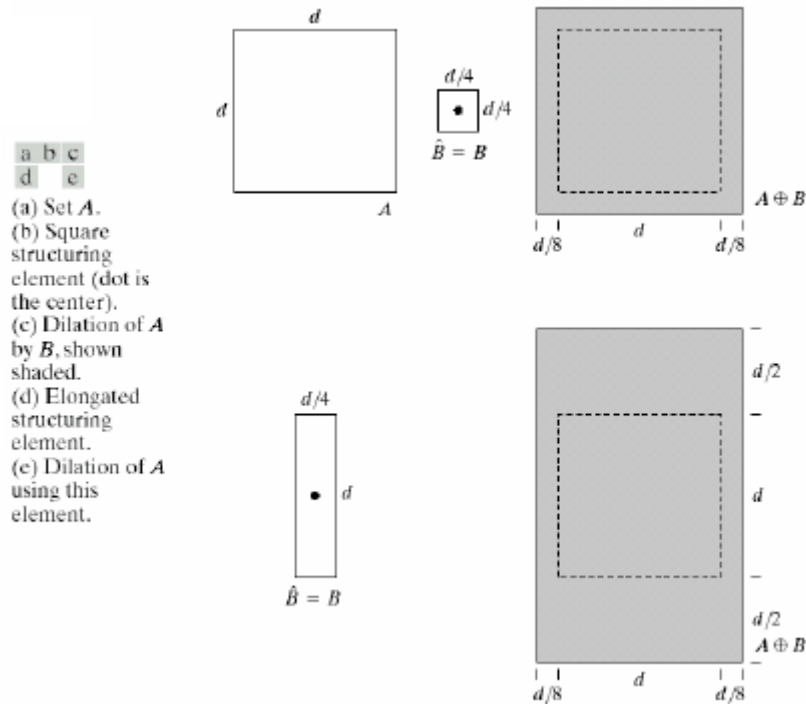
Q 22. Explain Dilation and Erosion.

Ans: **Dilation:**

The dilation of A by B is defined as –

$$A \oplus B = \left\{ z \mid \left(\hat{B} \right)_z \cap A \neq \emptyset \right\}$$

The dilation of A by B is the set of all displacements, such that $\hat{B}$ and A overlap by at least one element. Set B is commonly referred to as the **structuring element**.

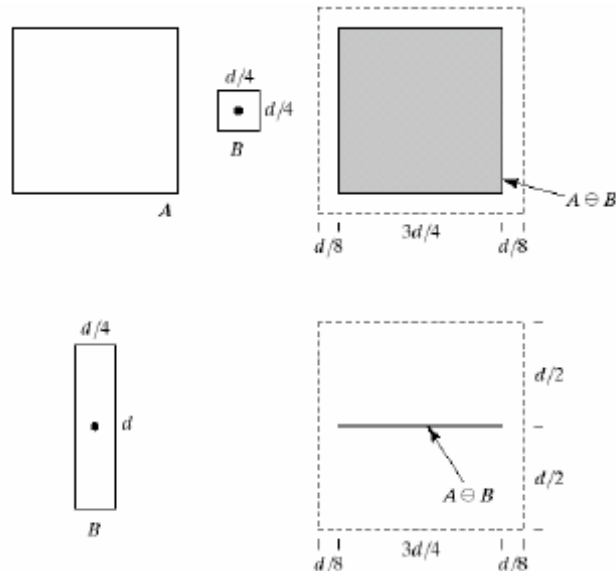


Erosion:

The erosion of A by B is defined as –

$$A \ominus B = \left\{ z \mid (B)_z \subseteq A \right\}$$

The erosion of A by B is the set of all points 'z' such that B, translated by 'z' is contained in A.



a b c
d e

(a) Set A. (b) Square structuring element. (c) Erosion of A by B, shown shaded. (d) Elongated structuring element. (e) Erosion of A using this element.

Q 23. Describe the techniques for edge and line detection.

Ans: Line Detection: Lines are characterized by abrupt change in intensity of gray levels of image. It also reflects a discontinuity in image. Following masks are used for line detection.

-1	-1	-1	-1	-1	2	1	2	-1	-1
2	2	2	-1	2	-1	-1	2	1	-1
1	1	1	2	-1	-1	-1	2	1	2
Horizontal			+45°			Vertical			-45°

By moving first mask on the image, we can detect the one pixel thick line in horizontal direction. The direction of line is simply decided by the coefficients of the mask. The preferred direction of the mask is weighted with larger coefficients than others.

Similarly, second, third and fourth masks can detect one pixel thick lines inclined at +45°, vertical line and lines inclined at – 45° respectively.

Edge Detection: Edge detection is also based on discontinuity measurement. It also shows the abrupt change in the intensity of gray levels. Edge can be defined as a set of connected pixels that lies in the boundary between two regions. An ideal edge is defined as a set of connected pixels each of which is located at an orthogonal step transition in gray level. Thus, an ideal edge has a step type response. But, practically edges can't be sharp. Practically, they are blurred and degree of blurring depend upon many factors such as quality of the image acquisition system, the sampling rate and the illumination conditions under which the image is acquired. As a result, edges are 'ramp like' profile.

Derivatives are used for measurement of edges or edge detection. If there is no edge, it means that there is no change in intensity levels, therefore, resultant derivative of image at that area will be zero. If the derivative is non – zero, there is a possibility of edges. If the first derivative is constant, edge is present. Now, take the second derivative. If it is positive, edge is at dark side and if it is negative, edge is at bright side.

Gradient and Laplacian are used for first and second order derivatives.

Q 24. Describe any one depth recover algorithm in detail.

Ans: Algorithm to recover path: Let us assume that the correspondence is established between the stereo image pair. Let, the disparity between two cameras is 2d. Disparity is the distance of separation between the left and right camera. Here, the two images of the point P(x, y, z) are formed at (x + d) and (x – d) location along x – axis.

If 'f' is the focal length of the pin – hole camera system and x_l and x_r are the left and right camera images respectively. Then,

$$x_l = \frac{(x - d)f}{f - z} \quad \text{and} \quad x_r = \frac{(x + d)f}{f - z}$$

$$z = f - \frac{2df}{x_r - x_l}$$

Thus, the depth of object could be recovered once the correspondence between both the images has been established

Q 25. Explain the process of image segmentation using region growing.

Ans: Region growing is a procedure that groups the pixels or sub-regions into larger regions based on predefined criteria. The basic approach is to start with a set of “seed” points and from these grow regions by appending to each seed those neighbouring pixels that have properties similar to the seed (such as specific ranges of gray level or colour). The selection of similarity criteria depends not only on the problem under consideration, but also on the type of image data available. Another problem in region growing is the formulation of a stopping rule.

The process can be defined as follows:

1. Start by choosing an arbitrary seed pixel and compare it with neighboring pixels.
2. Region is grown from seed pixel by adding in neighboring pixels that are similar, increasing the size of the region.
3. When the growth of one region stops, we simply choose another seed pixel which does not yet belong to any region and start again.

This is revised until all possible regions are found. This is called bottom up.

The main disadvantages of this technique are –

1. Current region dominates the growth process.
2. Different choices of seeds may give different segmentation results.
3. Problem can occur if arbitrary chosen seed point lies on the edge.

To remove these disadvantages, we use ‘simultaneous region growing’ technique.

1. Similarities between regions are also considered.
 2. At a time, a single region is not allowed to finalize. A number of parallel regions are grown.
 3. Process is difficult but many algorithms are available for easy implementation.
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